

7(a)	quantum of energy	M1
	of electromagnetic radiation	A1
7(b)(i)	photoelectric effect	B1
7(b)(ii)	- there is a frequency below which no electrons are emitted or threshold frequency = 5.4×10^{14} Hz - work function of the metal = 3.6×10^{-19} J (or 2.2 eV) E_{MAX} increases (linearly) with (increasing) frequency - gradient of the line is the Planck constant or gradient of the line is 6.7×10^{-34} J s <i>Any three bullet points, 1 mark each</i>	B3
7(c)(i)	different threshold frequency	B1
	(line has) same gradient but different intercept	B1
7(c)(ii)	photons have same energy	B1
	line unchanged	B1